

Research Assignment

# Title of the Research Assignment

Optional subtitle

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## 1 Introduction

Introduce the project objective and its context.

## 2 Methods

Describe the approach and implementation.

## 3 Results

Present and discuss the results.

## 4 Conclusion

Summarize the main outcome and remaining limitations.

## References

- [1] Kaiming He et al. “Deep residual learning for image recognition”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016, pp. 770–778.

## A LaTeX tips and examples

### A.1 General

- Use present tense in general: “For evaluation, we use the  $F_1$  score . . .” “The advantage of the method is its lower memory use . . .” and “We find that performance improves with more training data.” *etc.*
- Use past tense for completed experiments: “We trained the model for 100 epochs.”, “We collected 1000 samples from the dataset.” *etc.*
- Cite entries from `references.bib` by their key, *e.g.* `\cite{he2016deep}` produces [1]
- Use the macros from `definitions.tex` for abbreviations such as `\eg`, `\ie`, and `\etc`, and for the number sets `\R`, `\Z`, `\N`, `\Q`, and `\C`. For example, `\x \in \R^n` produces  $x \in \mathbb{R}^n$
- Add a `\label` to sections, equations, figures, and tables. Refer to them with `\cref`, which automatically inserts the object type such as Figure, Table, or Equation. For example, `\cref{eq:f1-score}` produces Equation (1) whereas `\cref{fig:example-box-plot}` produces Figure 1
- In general, refer to each figure and table in the text before it appears

### A.2 Equations

Use `$...$` for inline math, *e.g.* for the definition of the  $F_1$  score, see Equation (1). Use the `equation` environment for numbered equations:

$$F_1 = \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}}, \quad (1)$$

where TP, FP, and FN are the numbers of true positives, false positives, and false negatives.

### A.3 Figures

In general, use wide figures and place them at the top or bottom of a page. *E.g.* Figure 1 uses the full text width, while Figure 2 uses subfigures. Put captions below figures and briefly explain all panels, symbols, units, *etc.* Prefer vector formats such as PDF for plots, graphs, and diagrams (*e.g.* if they contain text) to avoid rasterization artifacts.

### A.4 Tables

Use `booktabs`, omit vertical lines, and keep decimal places consistent. For comparative results such as metrics or ablations, bold the best value in each column where appropriate, as in Table 1.

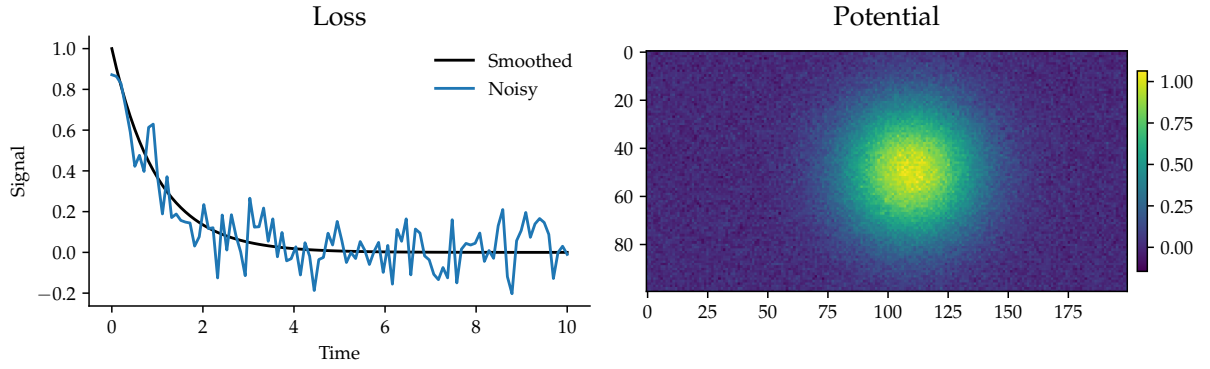


Figure 1: Example figure. Line plot showing a noisy signal and its smoothed trend (left). Heat map showing a spatial intensity distribution (right).

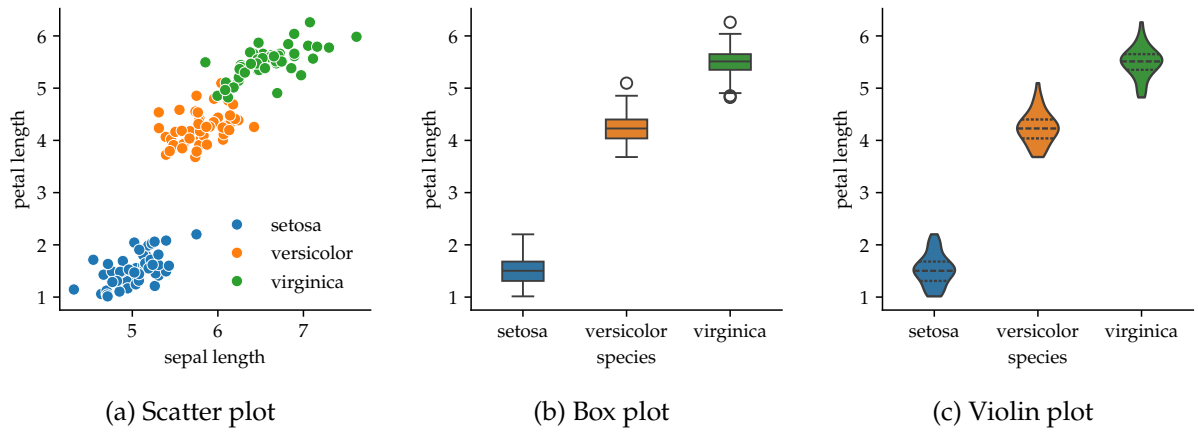


Figure 2: Alternative visualizations of the same data. Each panel shows the same synthetic Iris-style dataset.

| Method   | Accuracy $\uparrow$ | Precision $\uparrow$ | Recall $\uparrow$ | Error $\downarrow$ |
|----------|---------------------|----------------------|-------------------|--------------------|
| Method A | 0.81                | 0.80                 | 0.78              | 0.23               |
| Method B | 0.85                | <b>0.86</b>          | <b>0.84</b>       | 0.19               |
| Method C | <b>0.89</b>         | 0.84                 | 0.82              | <b>0.15</b>        |

Table 1: Example quantitative comparison. Shown are accuracy, precision, and recall (higher is better) as well as error (lower is better) for three methods.